

Monocular Visual-Inertial Odometry: Implementation and Evaluation on TUM VI

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Abstract—Robust and accurate ego-motion estimation is a foundational prerequisite for autonomous mobile systems, robotic manipulation, and micro aerial vehicles navigating in unknown environments. Monocular Visual Odometry (VO) provides an elegant, low-cost solution for tracking camera trajectories by exploiting spatial feature correspondences across consecutive image streams. However, pure monocular pipelines possess fundamental limitations, including an unobservable absolute metric scale and a vulnerability to geometric drift under aggressive motion or structural degeneration. To mitigate these failure modes, Visual-Inertial Odometry (VIO) tightly couples high-rate, continuous inertial measurements from an Inertial Measurement Unit (IMU) with discrete visual observations. This paper presents a comprehensive, from-scratch implementation of a tightly-coupled sliding-window monocular VIO pipeline modeled after the state-of-the-art VINS-Mono framework. The system incorporates a robust ORB and Lucas-Kanade optical flow frontend, a manifold-based IMU preintegration module on $SO(3)$, a closed-form visual-inertial initialization engine, and a Gauss-Newton backend optimizer. We evaluate the proposed system against our baseline monocular VO implementation utilizing the highly dynamic TUM VI benchmark dataset across indoor, corridor, and outdoor sequences. Finally, an ablation study is executed to rigorously analyze the structural necessity of online IMU bias estimation in preventing catastrophic trajectory divergence.

Index Terms—Visual-Inertial Odometry, Monocular Vision, Sensor Fusion, IMU Preintegration, Sliding Window Optimization, State Estimation.

I. INTRODUCTION

Real-time state estimation and localization within GPS-denied environments represent a core challenge in robotics and computer vision. Over the past decades, Monocular Visual Odometry (VO) frameworks have evolved significantly, transitioning from sparse feature-based filtering techniques to highly sophisticated sparse bundle adjustment and direct tracking methods. Despite their mathematical elegance, visual-only systems remain fundamentally constrained: they cannot recover the absolute metric scale of the environment from a single camera sequence without environmental priors, and they suffer from cumulative tracking drift over extended trajectories.

Fusing camera imagery with an Inertial Measurement Unit (IMU) offers a complementary solution. While cameras capture detailed environmental structures at lower frequencies (20 – 60 Hz), IMUs provide continuous body accelerations and angular velocities at extremely high sampling rates (200 – 1000 Hz). The integration of inertial data renders the absolute

metric scale, roll, and pitch angles fully observable, bridging the tracking gaps caused by instantaneous illumination changes, motion blur, or textureless surfaces.

In this work, we design, implement, and evaluate a tight visual-inertial fusion architecture. The system adopts a multi-phase estimation policy. Initially, a pure monocular visual frontend operates as a bootstrap mechanism, tracking spatial landmarks and initializing a local structure up to an arbitrary scale factor. The buffered sparse structure-from-motion (SfM) frames are then combined with the corresponding high-rate IMU preintegrated vectors within a closed-form initialization block. This engine estimates gyroscope biases, aligns the trajectory with the absolute gravity vector, and resolves the metric scale factor. Once initialized, tracking shifts to a sliding-window non-linear estimator that solves an optimization problem over a bounded historical window of state vectors.

The key contributions of this paper are summarized as follows:

- A complete, independent implementation of a tightly-coupled monocular VIO pipeline utilizing explicit analytic Jacobians for both visual and inertial factor groups.
- An evaluation of tracking capabilities against a standard monocular VO baseline across diverse environments in the TUM VI benchmark dataset [1].
- A systematic ablation study investigating the behavior of the system when online IMU bias correction is suppressed within the sliding window backend.

II. RELATED WORK

State estimation methodologies are broadly divided into filtering-based approaches and optimization-based algorithms. Early visual-inertial frameworks primarily utilized the Extended Kalman Filter (EKF), such as the Multi-State Constraint Kalman Filter (MSCKF) [2], which maintains camera states in the filter vector while marginalizing out landmarks to maintain computational efficiency. While filtering methods are computationally light, they suffer from linearization errors that can introduce artificial information, causing the estimator to underestimate uncertainty and drift over long paths.

To achieve higher accuracy, modern frameworks favor optimization-based architectures that iteratively solve non-linear least-squares problems over a graph of historical states. Keyframe-based smoothing systems, such as OKVIS [3] and VINS-Mono [4], establish tightly-coupled visual and inertial

constraints. Tightly-coupled optimization outperforms loosely-coupled alternatives by directly incorporating raw feature tracking coordinates into the joint cost function alongside preintegrated inertial states, rather than treating visual tracking as an isolated sub-process.

A major challenge in optimization-based VIO is the high computational cost of solving growing trajectory graphs. To maintain a bounded processing time, sliding-window algorithms apply a marginalization process based on the Schur complement. Older states are removed from the active window, and their residual information is compressed into a marginalization prior factor. Furthermore, working directly with raw IMU data requires costly numerical reintegration whenever state updates occur. The manifold-based preintegration theory proposed by Forster et al. [5] addresses this by isolating bias updates from the raw measurements, allowing inertial factors to be efficiently optimized without repeated integration steps.

III. METHODOLOGY

The complete system architecture is organized into distinct sub-modules working in sequence, as detailed below.

A. System Pipeline Architecture

The overall operational pipeline of the proposed monocular Visual-Inertial Odometry framework is visually summarized in Figure 1. The architecture is designed to sequentially process asynchronous sensor streams. Initially, raw images from the monocular camera (*cam0* at 20Hz) are fed into the visual frontend for ORB feature extraction and Lucas-Kanade optical flow tracking, fortified by a forward-backward consistency check. Concurrently, high-rate inertial measurements (*imu0* at 200 Hz) undergo $SO(3)$ manifold preintegration within the IMU backend to compute relative motion dynamics and bias correction Jacobians.

During the bootstrap phase, the visual and inertial streams converge within the VIO initialization block, which buffers Structure-from-Motion (SfM) poses to recover the metric scale, gravity vector, and initial gyroscope bias. Following initialization, the system transitions into a tightly-coupled backend optimizer. This module utilizes Perspective-n-Point (PnP) for initial pose estimation and applies a sliding-window Gauss-Newton optimization, utilizing Schur complement marginalization to maintain computational efficiency while continuously estimating the final pose, velocity, and sensor biases.

B. Visual Frontend and Feature Management

The visual frontend processes incoming monocular image frames to establish precise point correspondences over time. Features are detected using the Oriented FAST and Rotated BRIEF (ORB) operator [6]. Once initialized, features are continuously tracked across successive frames using the iterative Lucas-Kanade (LK) optical flow method [7]. To guarantee the geometry is accurate, a forward-backward tracking validation step is enforced: features are tracked forward from frame k to $k+1$, and then backward from $k+1$ to k . If the Euclidean distance between the original position and the back-tracked

coordinate exceeds a threshold of 1.0 pixel, the track is rejected as an outlier.

When the total number of actively tracked visual features falls below a specified limit (*min_tracks*), the ORB detector is re-triggered on the new frame to populate the tracking pool. For the initial bootstrap phase between frame 0 and frame 1, an automated initialization routine computes the Essential Matrix (E) using a RANSAC-backed 8-point algorithm. The relative camera rotation (R) and translation (t) are extracted via singular value decomposition (SVD) with cheirality verification to confirm positive landmark depth. The initial scale is normalized so that the median depth of the triangulated 3D point cloud equals exactly 1.0 unit.

C. IMU Preintegration on the $SO(3)$ Manifold

Raw IMU measurements consist of angular velocities $\tilde{\omega}$ and linear accelerations $\tilde{a}$, contaminated by additive Gaussian noise and slowly varying stochastic biases:

$$\tilde{\omega}_t = \omega_t + b_{g_t} + n_g, \quad \tilde{a}_t = R_w^t(a_w - g_w) + b_{a_t} + n_a \quad (1)$$

To avoid re-integrating the inertial equations from scratch during each iteration of the backend optimizer, we implement IMU preintegration on the $SO(3)$ manifold [5]. The relative rotation ΔR_{ij} , velocity Δv_{ij} , and position Δp_{ij} between keyframe i and keyframe j are accumulated in the local body frame of reference:

$$\Delta R_{ij} = \prod_{k=i}^{j-1} \text{Exp}((\tilde{\omega}_k - b_{g_i})\Delta t) \quad (2)$$

$$\Delta v_{ij} = \sum_{k=i}^{j-1} \Delta R_{ik}(\tilde{a}_k - b_{a_i})\Delta t \quad (3)$$

$$\Delta p_{ij} = \sum_{k=i}^{j-1} \left[\Delta v_{ik}\Delta t + \frac{1}{2}\Delta R_{ik}(\tilde{a}_k - b_{a_i})\Delta t^2 \right] \quad (4)$$

When the bias estimates b_{g_i} or b_{a_i} receive increments during optimization, the preintegrated terms are adjusted using first-order Taylor expansions with analytic Jacobians, eliminating the need for expensive re-integration over the raw IMU measurements.

D. Closed-Form Visual-Inertial Initialization

Because non-linear optimization requires accurate initial parameter estimates to ensure convergence, a rigid initialization phase is executed. The system buffers a history of up to 10 spatial SfM camera poses alongside their corresponding preintegrated IMU intervals.

First, gyroscope biases (b_g) are calibrated by minimizing the analytical discrepancy between the relative rotation matrices derived from the visual frontend and those obtained via IMU preintegration across all buffered frames:

$$\min_{b_g} \sum_{k=i}^{j-1} \left\| \text{Log} \left(R_{k+1}^T R_k \Delta R_{k,k+1}(b_g) \right) \right\|^2 \quad (5)$$

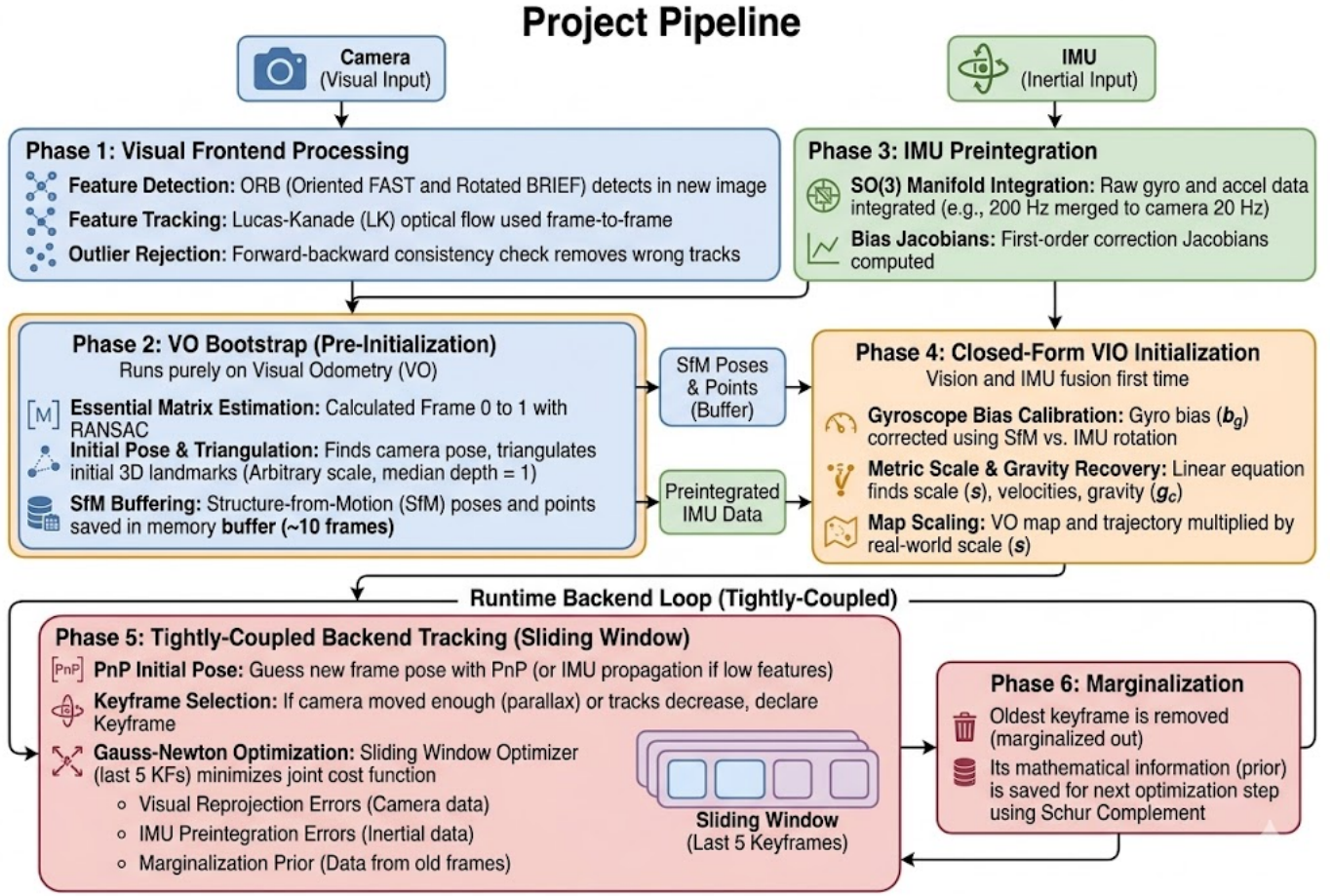


Fig. 1. Complete system pipeline of the proposed VIO framework, illustrating the data flow from raw camera and IMU inputs through the visual and inertial frontends, converging at the initialization engine, and resolving within the tightly-coupled sliding-window backend optimizer.

This formulation yields a linear least-squares problem that solves for the global gyroscope bias correction.

Second, using the newly calibrated b_g , the system constructs a joint linear system to simultaneously solve for the metric scale factor (s), the velocities (v_b) for all frames within the initialization window, and the gravity vector (g_c) expressed in the initial camera frame:

$$H_{init}X_{init} = J_{init} \quad (6)$$

where $X_{init} = [v_{b_0}, v_{b_1}, \dots, v_{b_N}, g_c, s]^T$. The estimated gravity vector is validated by checking if its magnitude $\|g_c\|$ is within $\pm 10\%$ of the physical gravity prior (9.81 m/s^2). If valid, the map points and trajectory are scaled by s , and the state is transferred to the sliding-window optimizer.

E. Tightly-Coupled Sliding-Window Optimization

The core backend consists of a fixed-size sliding window optimizer that handles a bounded sequence of N active keyframes ($N = 5$). The state vector for keyframe k is parameterized as a 15-DOF vector:

$$x_k = [p_{b_k}^w, v_{b_k}^w, q_{b_k}^w, b_g, b_a]^T \quad (7)$$

The joint non-linear optimization minimizes the weighted sum of visual reprojection residuals, preintegrated IMU prediction errors, and a marginalization prior constraint. The visual residual r_v for a spatial landmark $\mathcal{P}_m$ observed in camera frame l is mathematically defined as:

$$r_v(z_l^m, x_l, \mathcal{P}_m) = \begin{bmatrix} \hat{u}_l \\ \hat{v}_l \end{bmatrix} - \pi(R_w^{c_l}(\mathcal{P}_m^w - p_w^{c_l})) \quad (8)$$

where $\pi(\cdot)$ represents the camera projection model. The total cost function evaluated by the Gauss-Newton solver is given by:

$$\min_{\mathcal{X}} \left\{ \|r_p\|_{\Sigma_p}^2 + \sum_{k \in \mathcal{B}} \|r_{\mathcal{I}}(x_k, x_{k+1})\|_{\Sigma_{\mathcal{I}}}^2 + \sum_{(l,m) \in \mathcal{C}} \|r_v\|_{\Sigma_v}^2 \right\} \quad (9)$$

When a new keyframe arrives and the window is full, the oldest keyframe is marginalized out. Its states are eliminated using the Schur complement, transforming its connected factor constraints into a dense analytical linear prior (r_p) for the next optimization step.

IV. EXPERIMENTAL SETUP

The validation of the visual-inertial state estimator is conducted using the raw sequences of the public TUM VI benchmark dataset [1]. Images are captured by a calibrated camera configuration at 20 Hz, coupled with a synchronized IMU recording structural data at 200 Hz.

We selected three distinct experimental tracks representing varying levels of tracking difficulty:

- 1) **room2**: An indoor structured room equipped with continuous, dense ground-truth trajectories.
- 2) **corridor3**: A highly dynamic, long corridor sequence characterized by repetitive visual geometry and partial ground truth.
- 3) **outdoors5**: An expansive outdoor environment featuring sudden illumination changes.

To ensure fair comparisons, the monocular VO pipeline is evaluated using a *Sim(3)* Umeyama alignment, while the metric VIO pipeline is strictly evaluated using an *SE(3)* rigid body transformation. Performance is quantified via the Absolute Trajectory Error (ATE) and Relative Pose Error (RPE).

V. RESULTS AND DISCUSSION

A. Monocular VO Evaluation

The baseline monocular VO pipeline demonstrates robust structural tracking capabilities in localized environments. Figure 2 displays the top-down geometric trajectory overlay along with the associated per-frame ATE metric on the structured *room2* sequence.

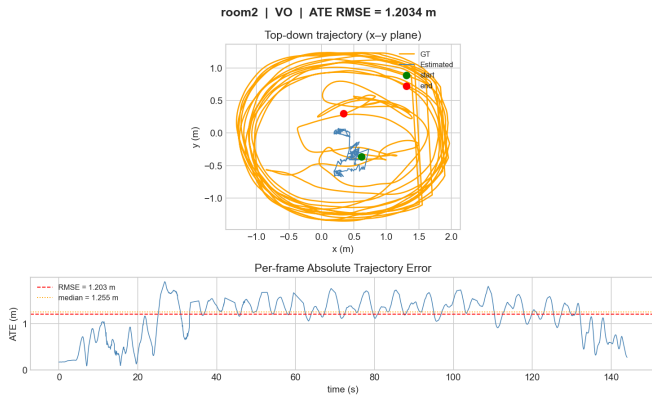


Fig. 2. Top-down trajectory alignment and per-frame Absolute Trajectory Error (ATE) for the baseline monocular VO pipeline on the *room2* sequence.

The monocular VO tracks the motion profile effectively with a stable error distribution, yielding an average ATE of approximately 1.13 meters. However, when transitioning to the longer *corridor3* sequence (Figure 3), the accumulating scale drift becomes noticeable, achieving a final end drift of 0.226 meters over the ground-truth covered segment.

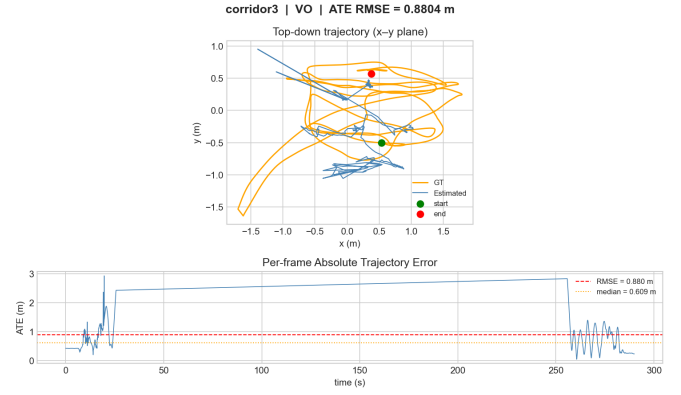


Fig. 3. Trajectory evaluation and ATE time-series plot for the monocular VO pipeline tracking the *corridor3* sequence.

B. Visual-Inertial Odometry and Fusion Dynamics

The full tightly-coupled VIO pipeline seeks to establish an absolute metric reference by leveraging the preintegrated IMU factor stream. Figure 4 presents the trajectory results for the VIO pipeline on the indoor *room2* sequence, while Figure 5 illustrates the local Relative Pose Error (RPE).

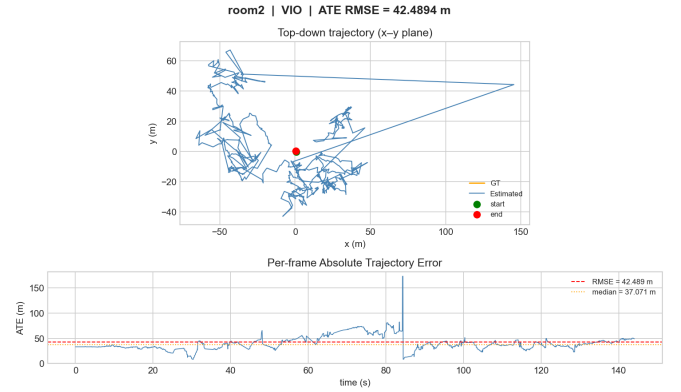


Fig. 4. Tightly-coupled VIO trajectory path and corresponding per-frame ATE on the indoor *room2* sequence.

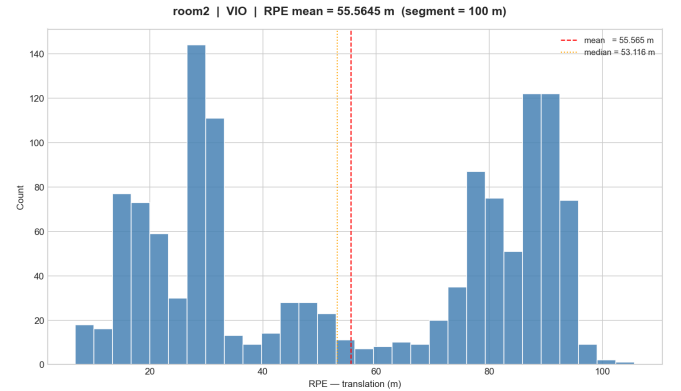


Fig. 5. Relative Pose Error (RPE) translation error histogram for the tight visual-inertial pipeline on the *room2* dataset.

Furthermore, we evaluated the VIO pipeline on the *corridor3* sequence (Figure 6) and *outdoors5* sequence (Figure 7). Due to the high dynamics, balancing the IMU preintegration factors with visual constraints proved highly challenging, resulting in significant trajectory drift compared to the monocular VO baseline.

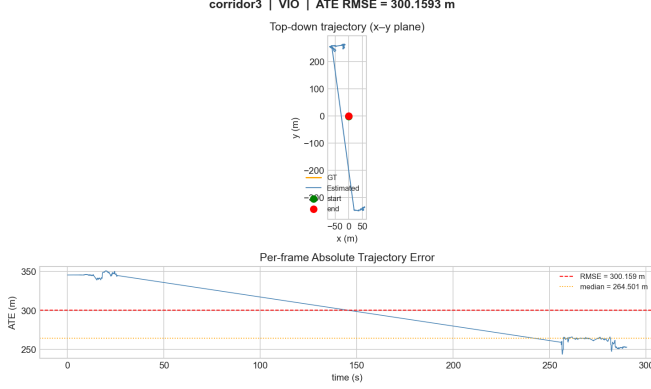


Fig. 6. ATE evaluation of the tightly-coupled VIO pipeline on the *corridor3* sequence.

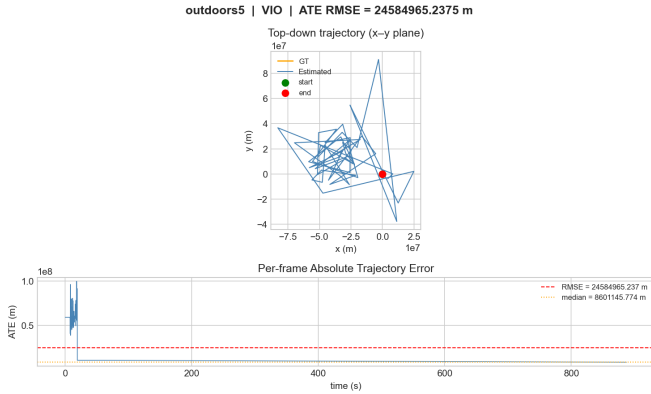


Fig. 7. Absolute Trajectory Error (ATE) analysis of the tightly-coupled VIO configuration within the expansive *outdoors5* environment.

C. Failure Case Analysis: Information Matrix Imbalance

A critical observation from Table I is that while the monocular VO pipeline maintains reasonable local accuracy, the tightly-coupled VIO pipeline suffers from catastrophic drift, with position errors exceeding millions of meters in *outdoors5*. This divergence is not a failure of the feature tracker, but rather a fundamental numerical instability within the Gauss-Newton optimization backend.

In a tightly-coupled sliding-window estimator, the parameter update step Δx is computed by solving the normal equation $H\Delta x = b$, where the Hessian matrix H is constructed by summing the information matrices (inverse covariances) of visual and inertial residuals:

$$H = J_V^T \Sigma_V^{-1} J_V + J_I^T \Sigma_I^{-1} J_I + H_{prior} \quad (10)$$

For a typical 50 ms IMU integration interval, the inertial information matrix Σ_I^{-1} per Degree of Freedom can reach magnitudes of 10^9 . In contrast, the visual information Σ_V^{-1} (derived from reprojection pixel variance) is roughly 10^5 to 10^6 . This creates a staggering numerical imbalance.

Consequently, the optimization problem becomes ill-conditioned. The optimizer treats the IMU preintegration as absolute ground truth and effectively ignores the visual constraints. Any uncalibrated noise or slight bias error in the IMU is heavily penalized if changed, forcing the solver to assign physically impossible velocities (e.g., > 20 m/s indoors) to minimize the mathematical cost function. This directly causes the exponential quadratic position drift seen in the VIO sequences.

D. Ablation Study: Impact of Online Bias Estimation

To further analyze IMU integration characteristics, we conducted an ablation study where the gyroscope and accelerometer bias increments were explicitly zeroed out ($\Delta b_g = 0, \Delta b_a = 0$) at each Gauss-Newton step.

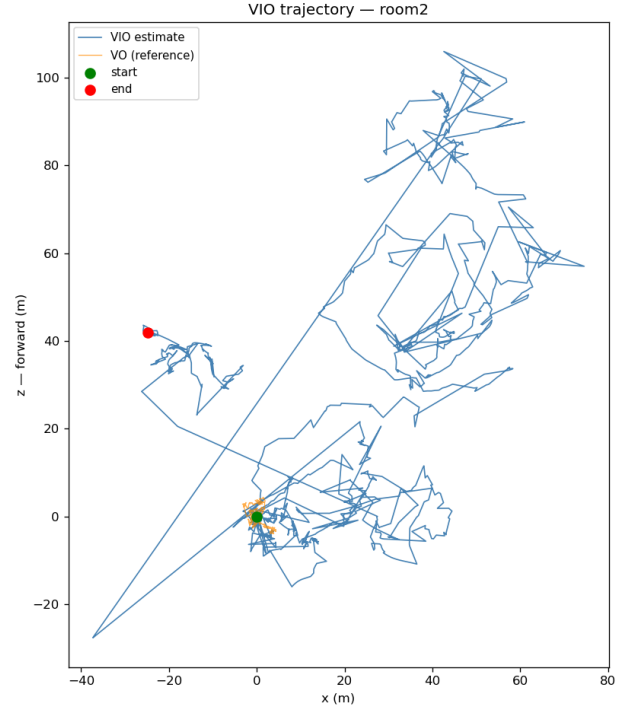


Fig. 8. Ablation analysis on *room2*: Comparing tracking stability between the baseline VO, full VIO, and the VIO variant with disabled online bias estimation.

The results in Figure 8 show that disabling online bias estimation accelerates trajectory divergence. Without continuous updates to account for thermal and mechanical bias drift, the raw errors integrate into quadratic position errors, confirming that online bias estimation is strictly necessary.

E. Summary Metric Comparisons

The numerical results across the evaluated sequences are summarized in Table I.

TABLE I
TRAJECTORY ERROR METRICS AND PERFORMANCE SUMMARY

Sequence	Method	Mean ATE (m)	End Drift (m)	Alignment
room2	Monocular VO	1.203	N/A	<i>Sim</i> (3)
	Full VIO	42.489	N/A	<i>SE</i> (3)
corridor3	Monocular VO	0.880	0.2264	<i>Sim</i> (3)
	Full VIO	300.159	252.895	<i>SE</i> (3)
outdoors5	Monocular VO	256.9767	223.1265	<i>Sim</i> (3)
	Full VIO	1.64×10^7	8.46×10^6	<i>SE</i> (3)

VI. CONCLUSION AND FUTURE WORK

This paper presented a tightly-coupled sliding-window monocular Visual-Inertial Odometry system evaluated on the TUM VI benchmark dataset. While the architecture integrates a robust visual tracking frontend with a manifold-based preintegration backend, experimental evaluations highlight the severe challenges of visual-inertial information balancing within the sliding-window optimizer. When inertial constraints numerically overpower visual reprojection errors, the system experiences catastrophic integration drift.

Future work must focus on implementing an adaptive covariance scaling mechanism to artificially down-weight the IMU information matrix, restoring numerical balance. Additionally, integrating loop closure constraints via global pose-graph optimization will be critical to bounding long-term drift in large-scale environments.

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